6.Patient Appointment Automation Suite — Workflow Options Use Case

Aim:

To design and document the automation layer in Zoho Creator for the **"Patient Appointment & Scheduling Management"** application, demonstrating the configuration of seven workflow categories: Form Workflows, Schedules, Approvals, Blueprints, Batch Workflows, Report Workflows, and Functions. These automate appointment validation and confirmation, reminders, special-procedure decisions, lifecycle stages, report-level actions, and visit-summary emails.

Workflow Options Used Overview

Workflow Option	Workflow Name	Purpose	Trigger / Frequency

Form Workflows	`Validate_Appointment`
	`Send Appointment Confirmation`
	`Send_Visit_Summary_On_Consulted`
		Validates booking requests, sends confirmation, and invokes a visit-summary action when consultation status is updated.	Created: validation and successful submission; Edited: Status update. All 3 enabled.

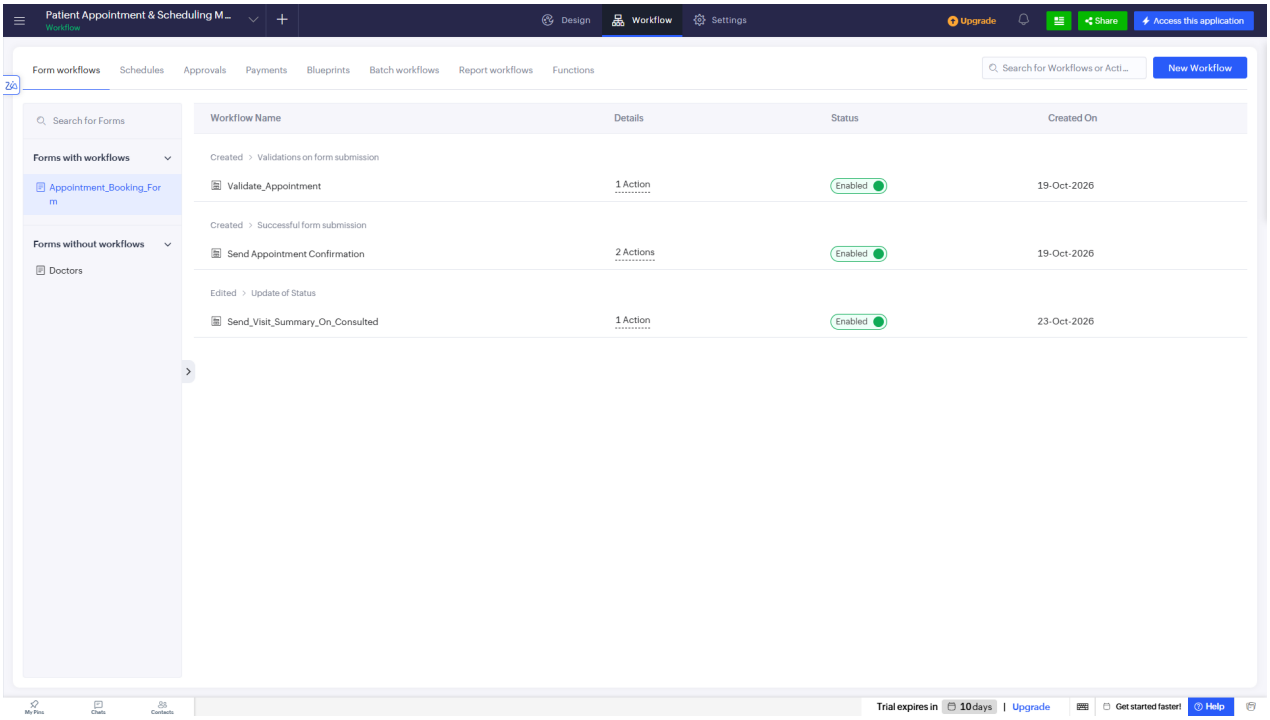
Schedules	`Daily_Appointment_Reminder_Schedule`	Emails eligible confirmed patients and marks the tomorrow-reminder flag after the reminder action.	Based on Preferred Appointment Date; two conditional actions; enabled.
Approvals	`Special_Procedure_Approval`	Routes special procedure requests through a Level 1 approval or rejection path with notifications and record updates.	On review when Special_Procedure_Requested = "Yes"; 5 actions; enabled.
Blueprints	`Appointment_Lifecycle_Blueprint`	Organizes appointment progress through request, confirmation, check-in, consultation, and cancellation stages.	Applied to Appointment_Booking_Form; enabled; stage transitions are shown in the editor.
Batch Workflows	`Bulk_FollowUp_Reminder_Batch`	Provides an enabled batch workflow entry for appointment follow-up reminders.	Scheduled trigger shown: 24-Oct-2026 21:00:00; enabled.

Workflow Option	Workflow Name	Purpose	Trigger / Frequency
Report Workflows	`OnEdit_Recalculate_NetFee_Workflow` `Resend_Visit_Summary`	Supports net-fee recalculation and resending a visit summary from report actions.	User trigger in a report; one action each; both enabled.
Functions	`Generate_VisitSummary_Function` `Send_Today_Appointment_Reminders`	Sends PDF visit summaries for eligible Consulted records and email reminders for today's Confirmed bookings.	Deluge functions with void return type; execution depends on their invocation.

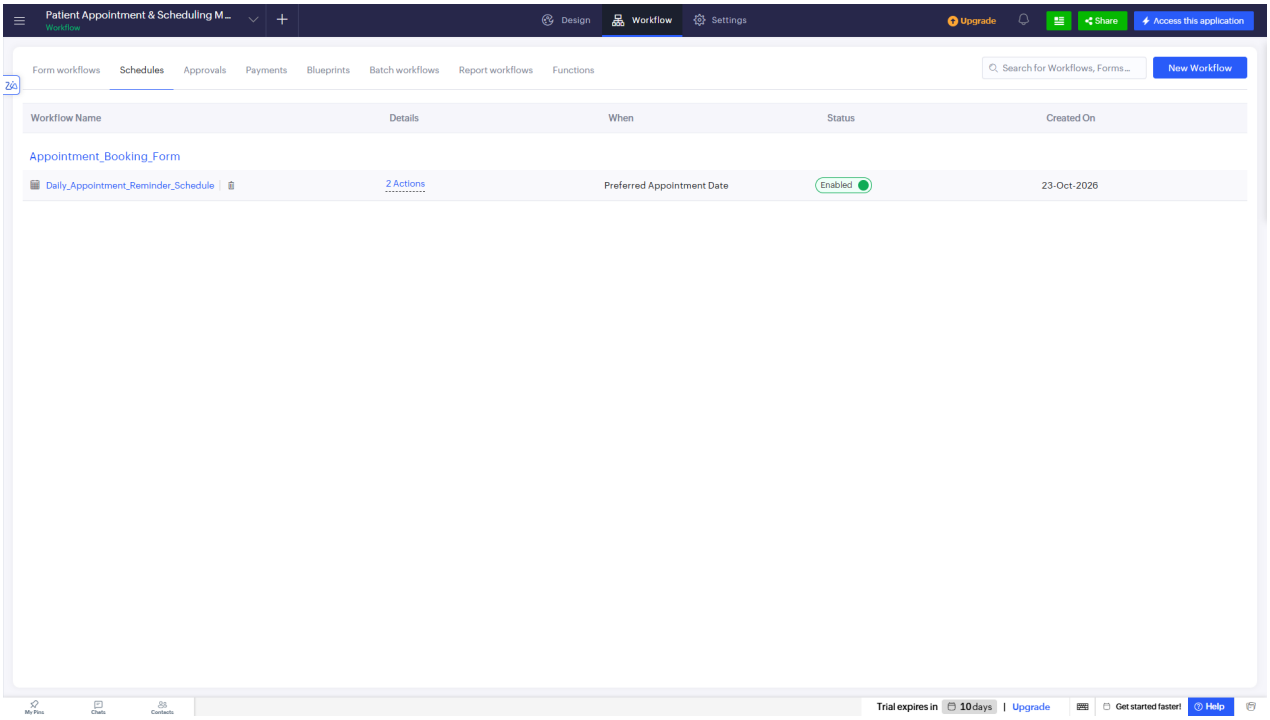
Automation Details & Workflow Breakdown

1. **Form Workflows: Validate_Appointment** is an enabled validation-on-submission workflow (1 action). **Send Appointment Confirmation** runs on successful form submission (2 actions). **Send_Visit_Summary_On_Consulted** is enabled for edits to Status (1 action).
2. **Schedules: Daily_Appointment_Reminder_Schedule** is enabled and uses **Preferred Appointment Date**. Its two actions show a conditional email to **Email_Address** and a Deluge script updating **Tomorrow_Reminder_Sent = true** after fetching the appointment. The visible condition checks Confirmed status, a non-null email address, and a reminder-related field; an exact execution hour is not shown.
3. **Approvals: Special_Procedure_Approval** is enabled for **Appointment_Booking_Form** and is conditional on **Special_Procedure_Requested == "Yes"**. A Level 1 approver has **Approve** and **Reject** paths; the screenshots show email notification to the submitter and record-update actions in the respective branches.
4. **Blueprints: Appointment_Lifecycle_Blueprint** is enabled for the booking form. The editor displays the Requested and Cancelled states and appointment progression via **Confirm**, **Check in**, **Complete Consultation**, and **Cancel** transitions. The screenshots establish the configured visual flow, not completed transitions on sample records.
5. **Batch Workflows: Bulk_FollowUp_Reminder_Batch** is **Enabled**. Its listed trigger is **24-Oct-2026 21:00:00**. This verifies the updated configuration, but no successful batch-run log is supplied.
6. **Report Workflows: OnEdit_Recalculate_NetFee_Workflow** and **Resend_Visit_Summary** are both enabled. Each has one action with the event **User trigger in a report**; the screenshots do not confirm automatic execution on every edit.
7. **Functions: Generate_VisitSummary_Function(int appointment_id)** checks for an appointment record and requires a patient name, doctor lookup, valid email, and **Status == "Consulted"**. Its Deluge code uses **sendmail** to attach a visit-summary template as a PDF. **Send_Today_Appointment_Reminders()** loops through bookings where **Preferred_Appointment_Date == zoho.currentdate** and **Status == "Confirmed"**, then emails patients with a non-null address to arrive 15 minutes early.

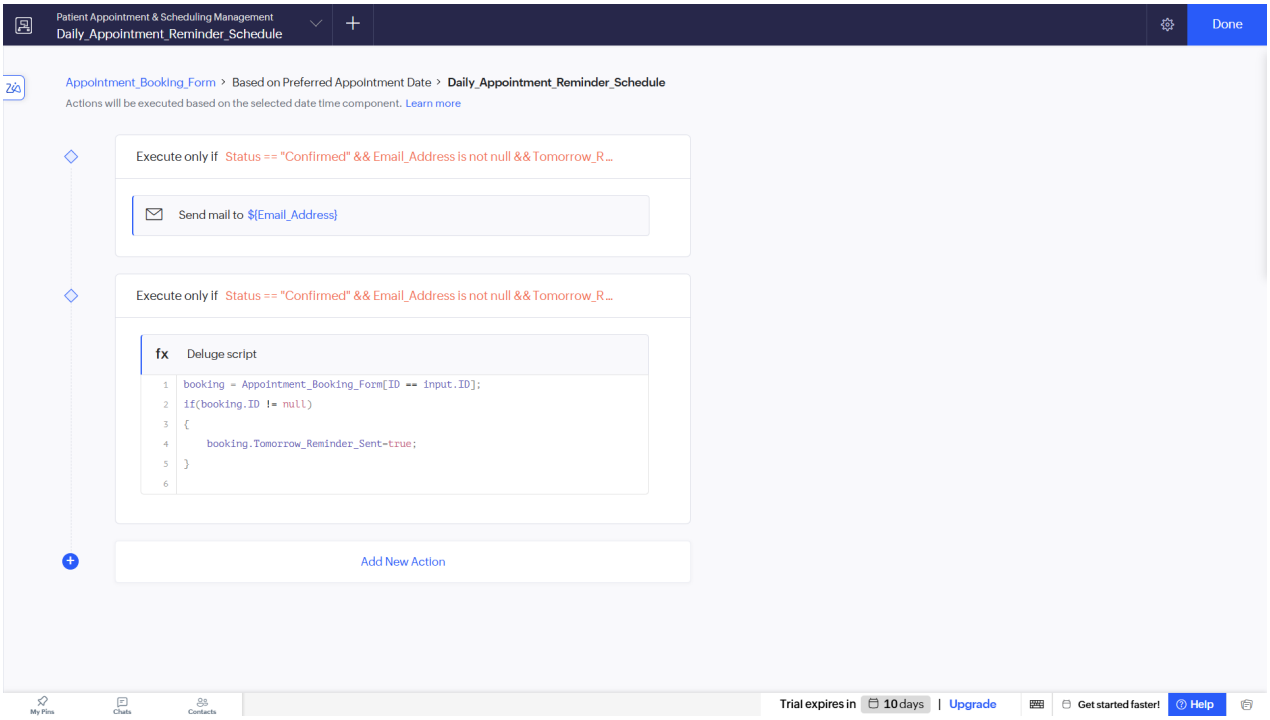
Screenshots:



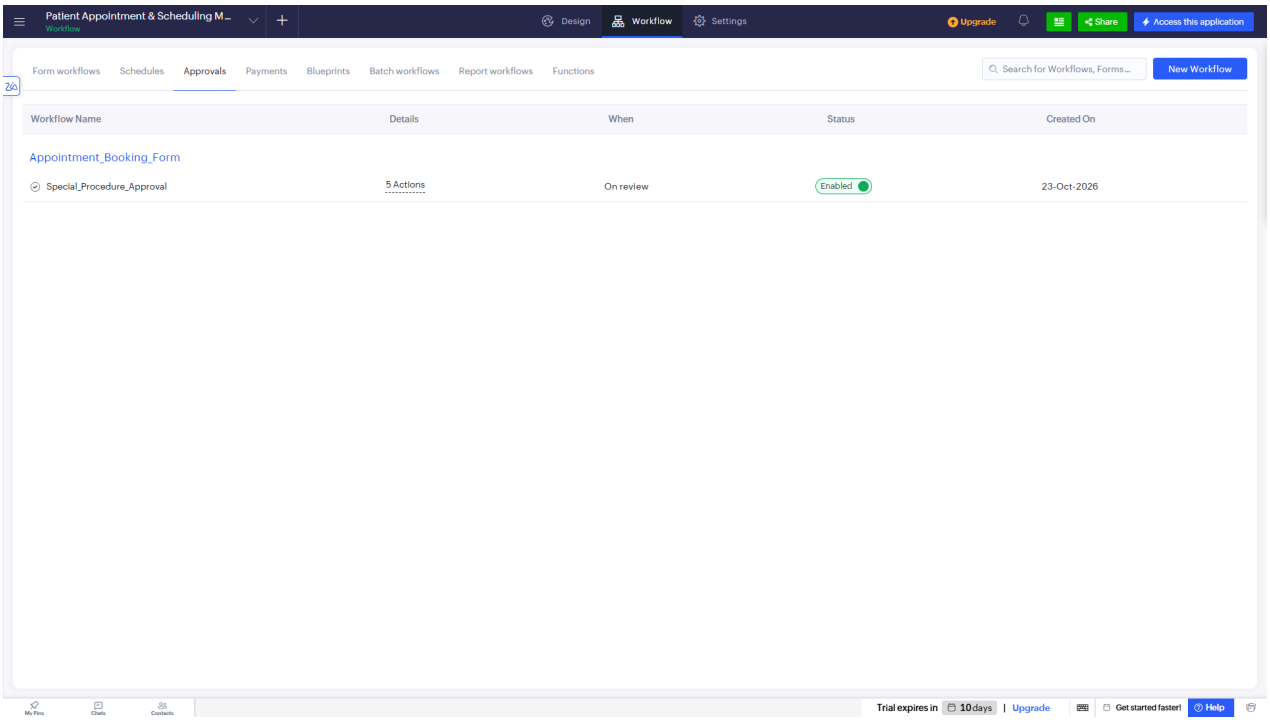
Screenshot 1: Form Workflows. Enabled form workflows for appointment validation, confirmation, and post-consultation summary.



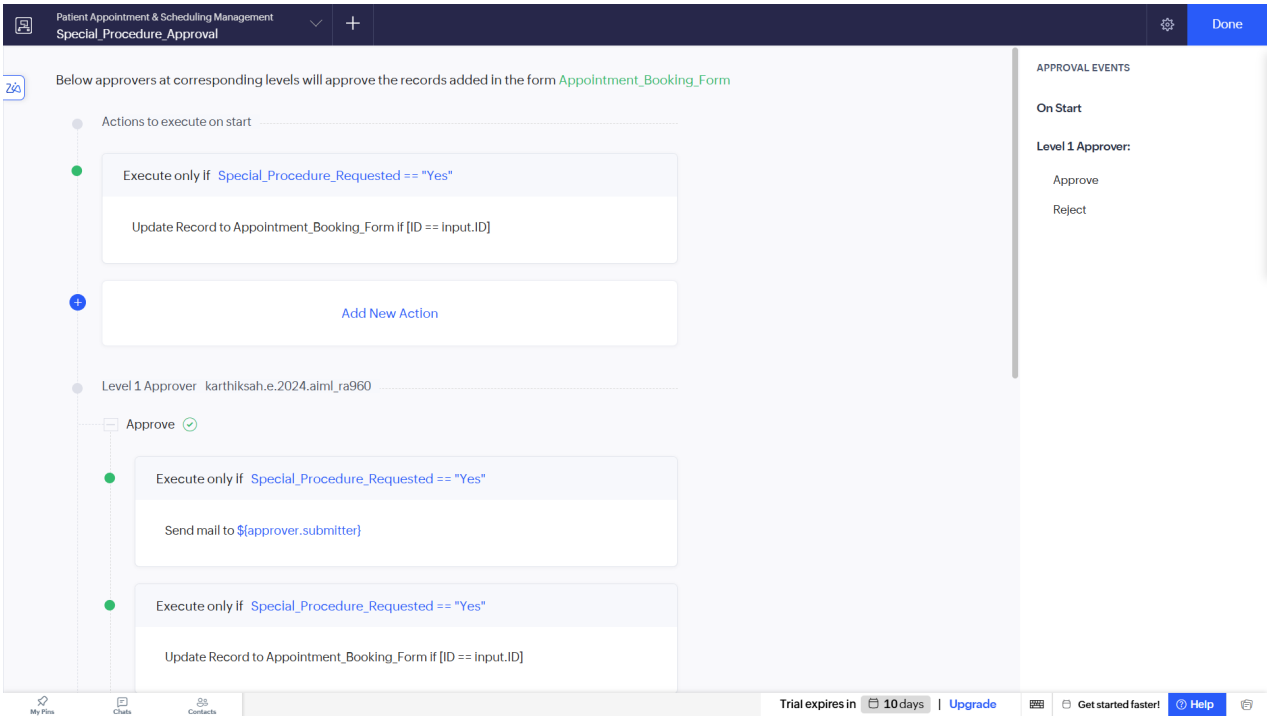
Screenshot 2: Schedules Overview. Enabled Daily_Appointment_Reminder_Schedule linked to Preferred Appointment Date.



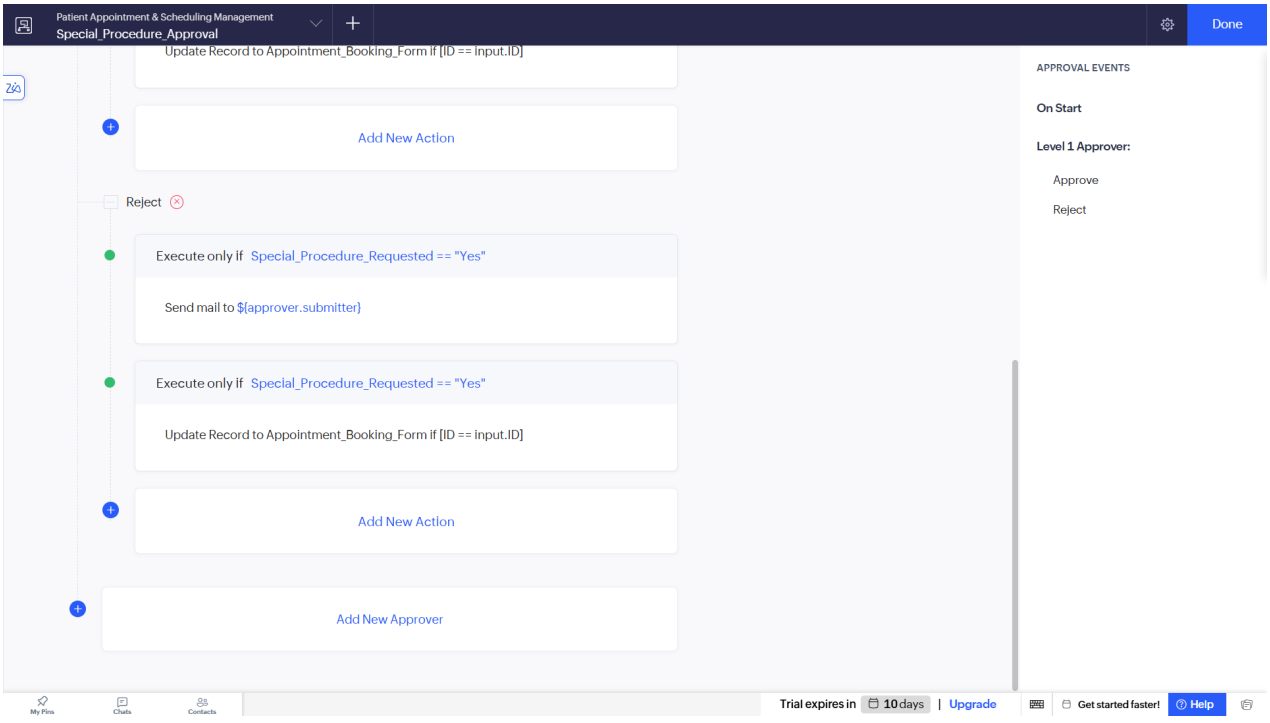
Screenshot 3: Schedule Actions. Conditional reminder email and a Deluge action setting Tomorrow_Reminder_Sent to true.



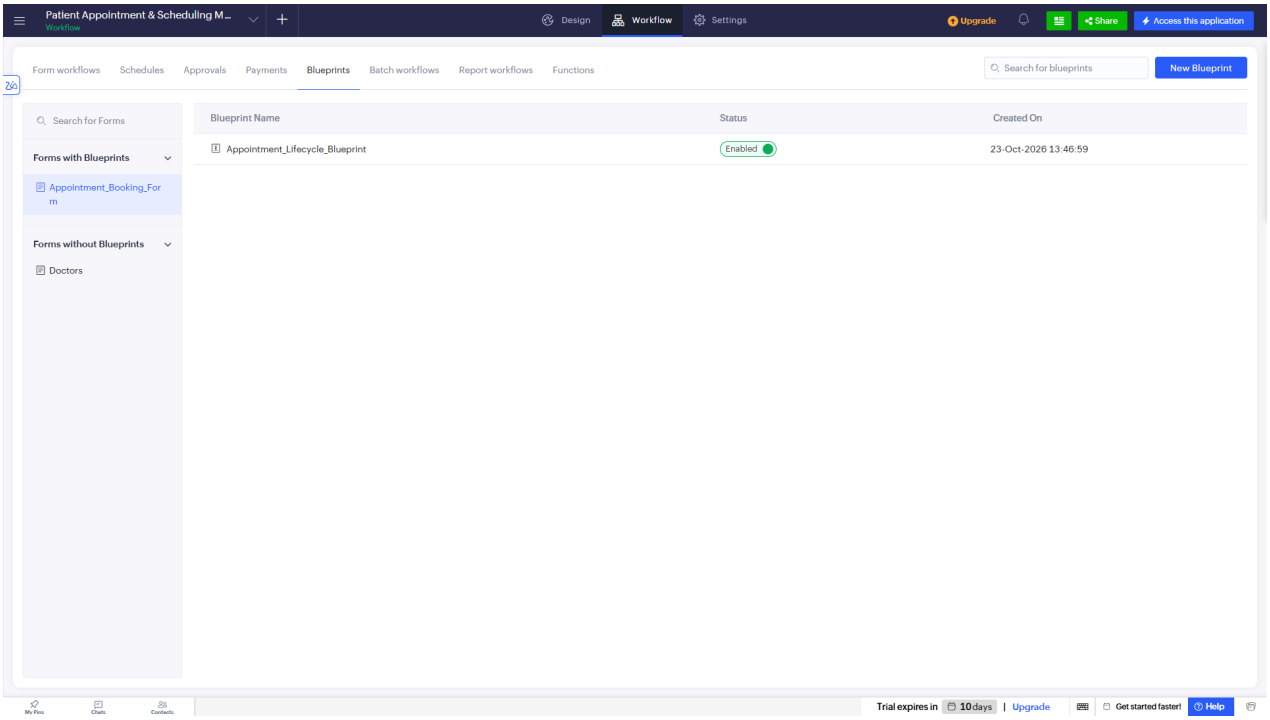
Screenshot 4: Approvals Overview. Enabled Special_Procedure_Approval with five configured actions.



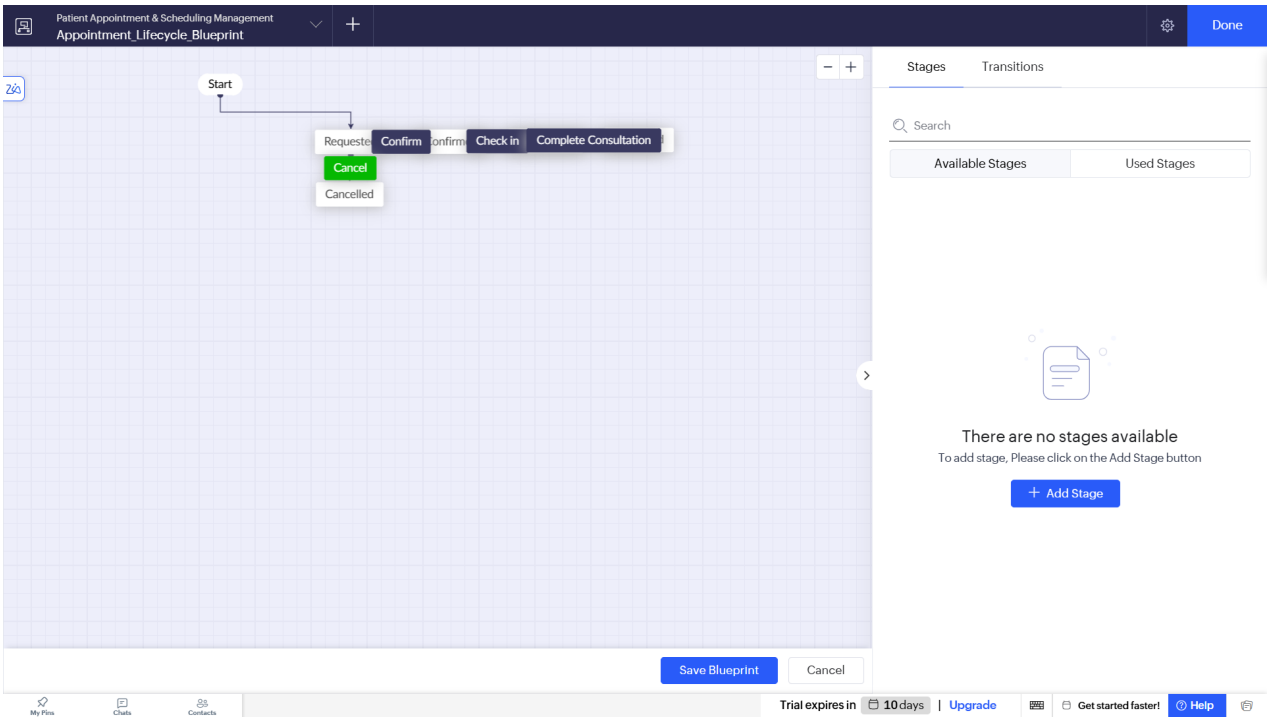
Screenshot 5: Approval - Start and Approve. Conditional actions for a special procedure request, including record updates and an email on approval.



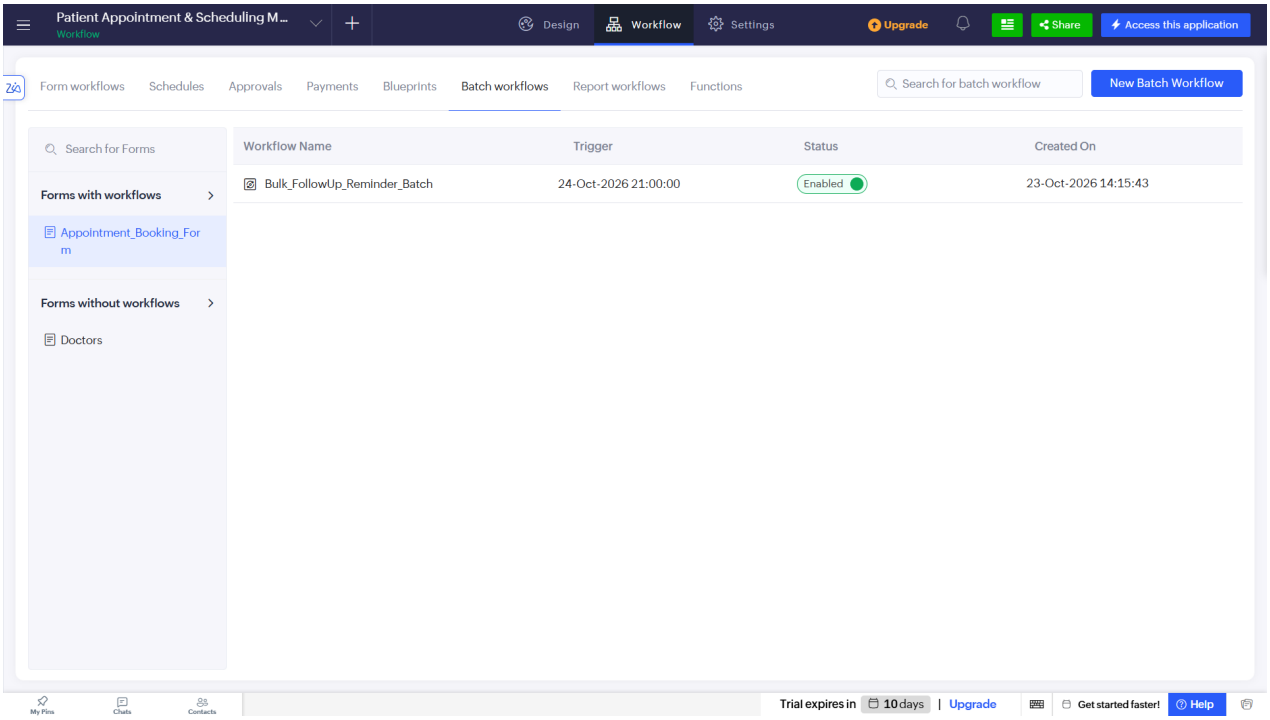
Screenshot 6: Approval - Reject. Rejection branch with notification and record-update actions.



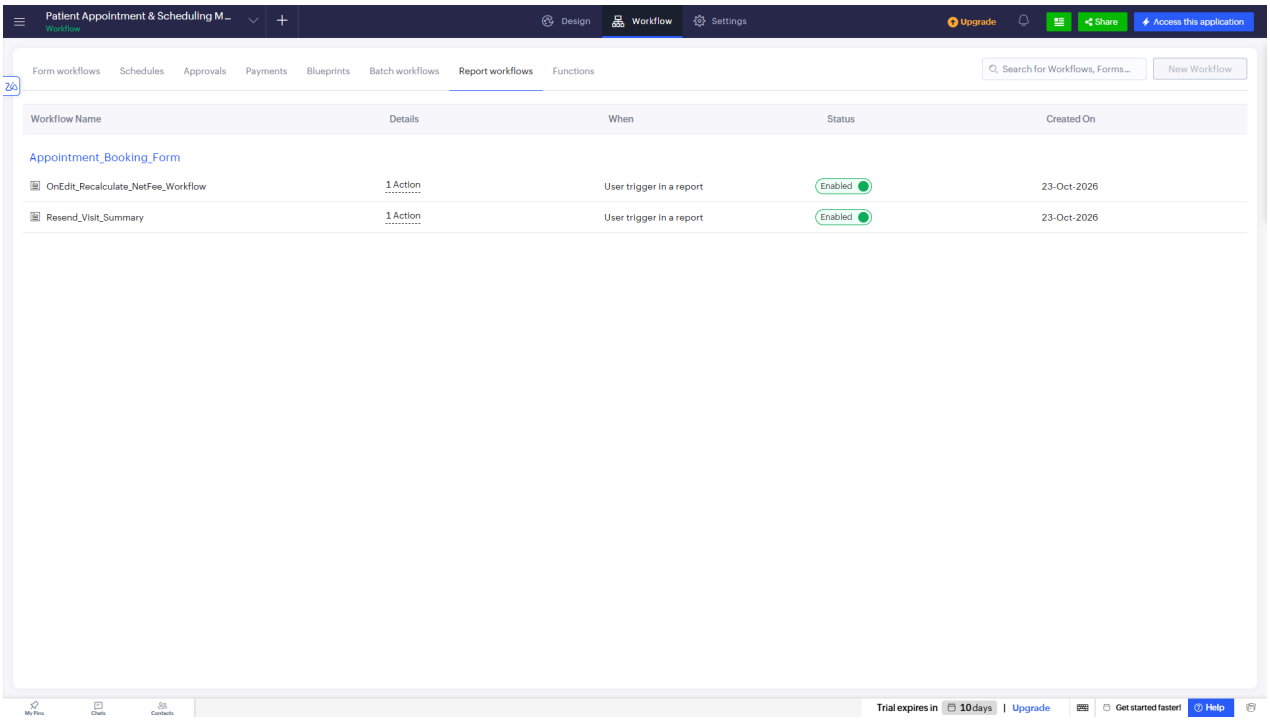
Screenshot 7: Blueprints Overview. Enabled Appointment_Lifecycle_Blueprint for the booking form.



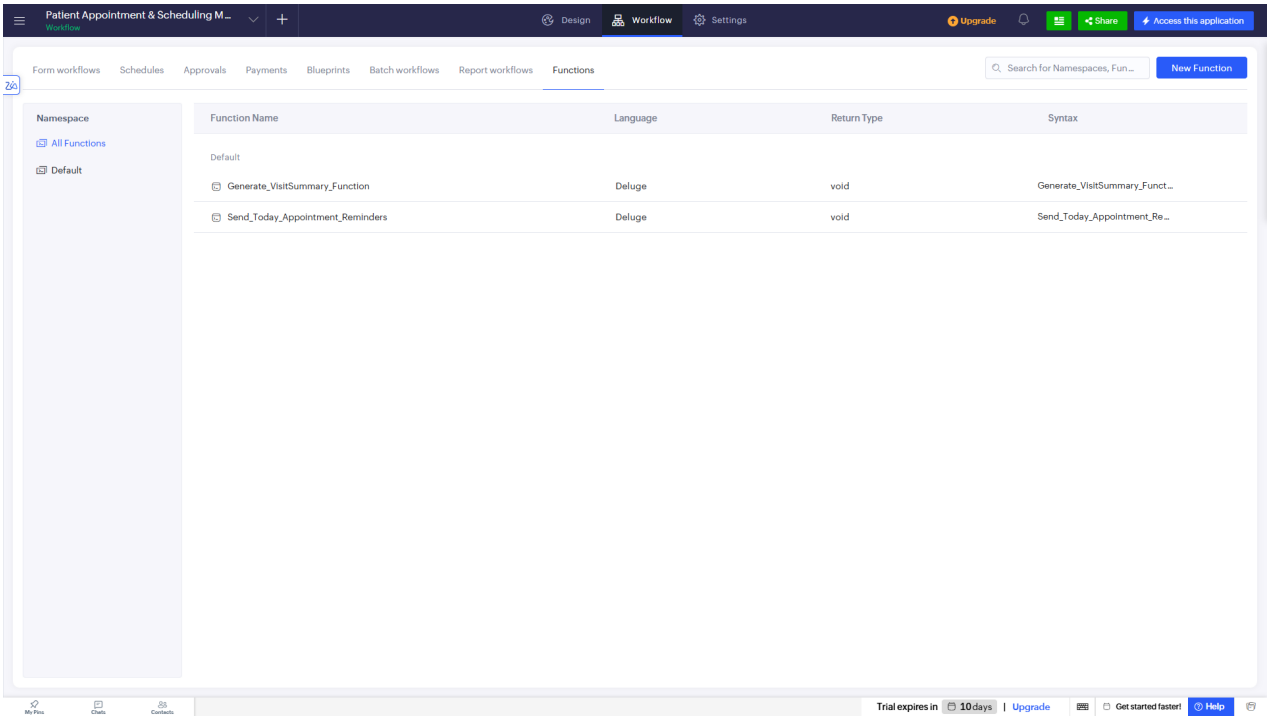
Screenshot 8: Appointment Lifecycle Blueprint. Visible lifecycle stages and transitions, including Confirm, Check in, Complete Consultation, and Cancel.



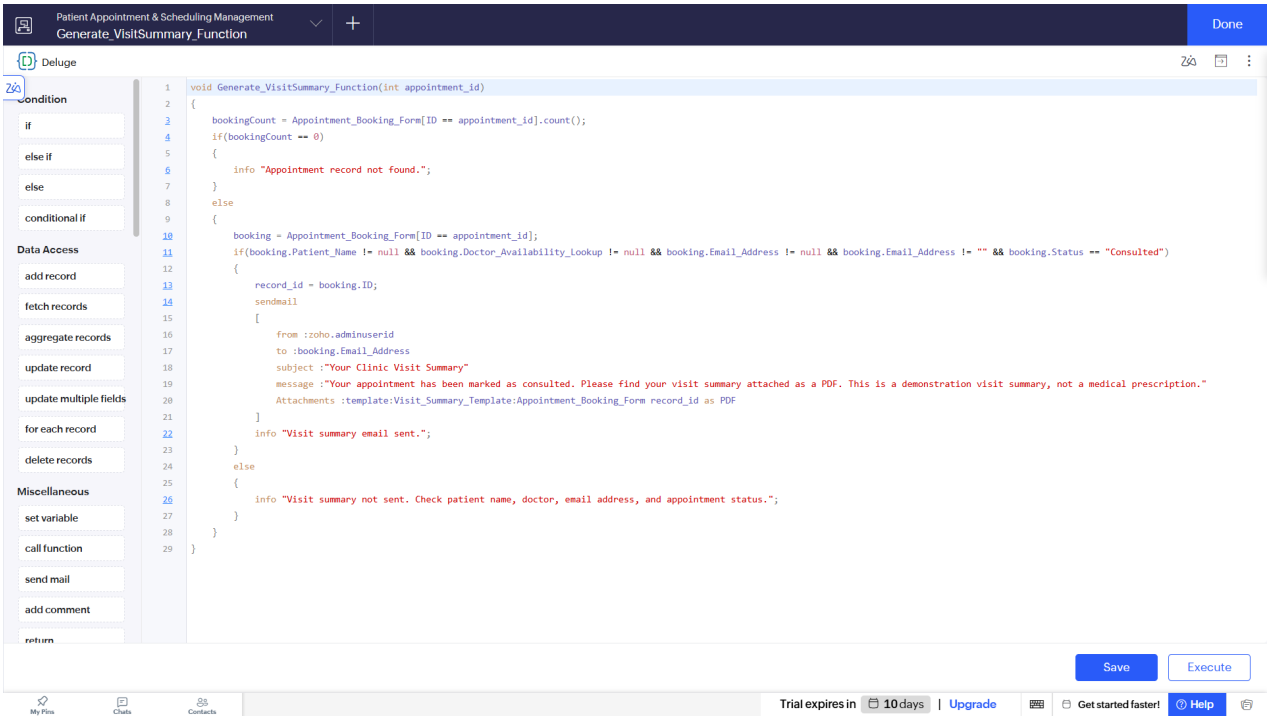
Screenshot 9: Batch Workflows. Bulk_FollowUp_Reminder_Batch is Enabled; trigger is 24-Oct-2026 at 21:00.



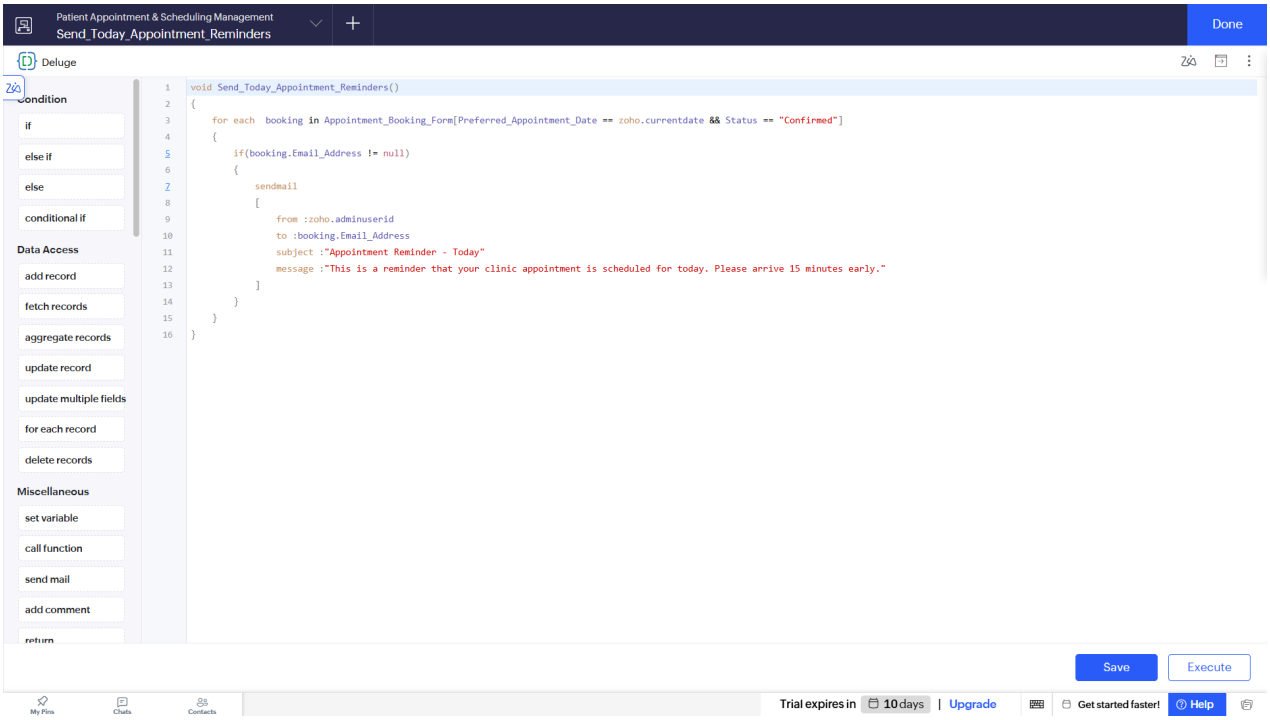
Screenshot 10: Report Workflows. Enabled net-fee recalculation and visit-summary resend actions triggered from reports.



Screenshot 11: Functions Overview. Generate_VisitSummary_Function and Send_Today_Appointment_Reminders listed as Deluge functions.



Screenshot 12: Visit-Summary Function. Deluge code checks the booking and sends a PDF visit summary for eligible Consulted records.



Screenshot 13: Today-Reminder Function. Deluge code selects today's Confirmed bookings and sends email reminders.

Result:

The **Patient Appointment & Scheduling Management** application has a documented automation suite covering **seven configured workflow categories**. The screenshots confirm enabled form workflows, schedules, approvals, blueprint, batch workflow, and report workflows, together with the Deluge functions for visit summaries and same-day reminders. The **Bulk_FollowUp_Reminder_Batch** is **Enabled** with a trigger of **24-Oct-2026 21:00:00**. These screenshots demonstrate configuration; successful delivery or execution is not independently verified.